LinkedIn DM Lead Tracker: Process and Outcome

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The problem

Salespeople message leads on LinkedIn, but the CRM never hears what happened next. Did the lead open it? Reply? Say yes or no? Who needs a nudge today?

The goal: keep each lead's status in the CRM current (message sent, opened, replied, not replied) without changing how salespeople work. They use LinkedIn as usual, and the CRM updates itself.

Why the official LinkedIn API does not work

LinkedIn's messaging API is limited to approved partners, with no public way to apply. Nobody can read DMs through it on their own, so a working solution has to read the inbox some other way.

Options compared

Option	Opened	Replied	Account risk	Verdict
Unipile API	Yes	Yes, real time	Medium	Built first, then dropped
Chrome extension reading our own inbox	Yes, via the Seen marker	Yes	Lower	**Chosen**
LinkedIn notification emails in Gmail	No	Mostly	None	Zero risk fallback
Outreach tools (HeyReach, Expandi) plus Zapier	Yes	Yes	Medium	Configuring, not building
Headless Playwright bot	Yes	Yes	High	Ban risk

The pivot

I first built the planned version on Unipile, a service that connects a LinkedIn account and sends webhooks: a Node server, a SQLite event store, the status engine, a Grok classifier and a Sheets adapter, with 98 tests (kept under the `unipile-version` git tag). Two things changed my mind. The Unipile account was suspended soon after first use, which is exactly the risk the brief warned about. And setup needed about 11 keys, ngrok and an always-on server.

So I moved to the backup plan: a Chrome extension that reads the LinkedIn Messages page the user already has open, and a Google Apps Script inside the Sheet as the backend. No server, no service account, and setup is two values: the script's web app URL and the AI key. The status engine carried over as is.

Architecture

```
LinkedIn Messages page (open in Chrome)
   |  extension reads names, previews, the open thread and the Seen marker
   v
Chrome extension  --POST JSON-->  Google Apps Script web app (inside the Sheet)
                                    |- observe()    compares the screen with the row
                                    |- nextState()  pure status rules, unit tested
                                    |- CRM adapter  Leads, Events, Summary tabs
                                    |- classifier   Grok labels replies, drafts follow-ups
                                    '- timer        Follow-up due / Not replied
```

Choices worth calling out:

- **Compare, don't count.** The extension sends what it sees, not events. The script compares it with the row, and only a real difference becomes an event, so sending the same screen twice changes nothing.
- **One clock.** LinkedIn's "2:15 PM" labels are too rough to order messages, so every event uses the time the extension saw it.
- **Tracking starts at install.** Old chats become the starting point, not fake replies, so an old inbox never floods the Sheet.
- **The AI never blocks a status change.** The row moves to Replied first. If Grok fails, the category is Unclassified and everything carries on.
- **Humans send.** The extension never clicks, types or sends. Drafts go to the Suggested Follow-up column.

Checked against the real page. I first wrote the page selectors from memory, then checked them on the live Messages page. Two guesses were wrong: the conversation list has no links (so the extension learns each thread's id

when it is opened and remembers it by name), and a loosely matched "seen" class also matched an element that is always on the page. Messages with a subject line preview as the subject, so the open thread decides who sent them.

Status rules

Status always reflects the latest turn, and never moves backwards unless we send a new message.

Trigger	Condition	New status
Lead row added	none	Not contacted
New message from us	any	Message sent
Seen marker on our latest message	status is Message sent	Opened
New message from the lead	any	Replied, plus AI label
Timer, no reply after N days	Message sent or Opened	Follow-up due, or Not replied once follow-ups run out

Defaults are 3 days and 2 follow-ups; demo mode reads days as minutes. LinkedIn's web page has no delivered marker, so the brief's Delivered status was dropped rather than kept as a step that never happens.

Known limitations

- **It only syncs while LinkedIn Messages is open in Chrome.** Messages that arrive while it is closed are picked up later, timed when they were seen.
- **A conversation is tracked once it has been opened since install**, because only the open thread shows its id. Messaging a lead opens their thread, so this mostly affects old chats.
- **Opened needs the lead's read receipts on**, and the thread open while the Seen marker shows. Otherwise the lead stays at Message sent until they reply.
- **It depends on LinkedIn's page layout.** All selectors live in one file, and the popup's "Check this page" shows which still match.
- **Lower risk, not zero.** Reading your own screen is still scraping in the strict sense, though it makes no requests to LinkedIn.
- **The web app URL works like a password**, and Google Sheets is not a CRM at scale (Apps Script quotas).

Path to HubSpot or Zoho

The rest of the script only talks to four methods: `listLeads`, `upsertLead`, `updateStatus` and `logEvent`. For HubSpot, leads become contacts with a few custom properties (tracker id, DM status, reply category, timestamps), `logEvent` becomes a note on the contact, and calls go through `UrlFetchApp` with a private app token. Zoho maps the same way onto its Leads module. The README has the method by method mapping.

Outcome

- A Chrome extension and an Apps Script that turn LinkedIn DM activity into live CRM status, with AI labels on replies and drafted follow-ups.
- 226 automated tests: status rules, matching, the Sheets adapter, the AI fallback, the web app, the extension's parser on LinkedIn-shaped HTML, and an end-to-end run from page to Sheet row.
- A live Sheet (Leads, Events, Summary) and a demo video of a row going from Message sent to Opened to Replied, and a quiet lead flipping to Follow-up due.